Fish and Shellfish Immunology 152 (2024) 109785

Contents lists available at ScienceDirect

Fish and Shellfish Immunology

journal homepage: www.elsevier.com/locate/fsi

The mucosal immunity in crustaceans: Inferences from other species

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ARTICLE INFO

Keywords:
Crustaceans
Mucosal immunity
Immune-modulation
Innate immunity
Comparative immunology

ABSTRACT

Crustaceans such as shrimps and crabs, hold significant ecological significance and substantial economic value within marine ecosystems. However, their susceptibility to disease outbreaks and pathogenic infections has posed major challenges to production in recent decades. As invertebrate, crustaceans primarily rely on their innate immune system for defense, lacking the adaptive immune system found in vertebrates. Mucosal immunity, acting as the frontline defense against a myriad of pathogenic microorganisms, is a crucial aspect of their immune repertoire. This review synthesizes insights from comparative immunology, highlighting parallels between mucosal immunity in vertebrates and innate immune mechanisms in invertebrates. Despite lacking classical adaptive immunity, invertebrates, including crustaceans, exhibit immune memory and rely on inherent "innate immunity factors" to combat invading pathogens. Drawing on parallels from mammalian and piscine systems, this paper meticulously explores the complex role of mucosal immunity in regulating immune responses in crustaceans. Through the extrapolation from well-studied models like mammals and fish, this review infers the potential mechanisms of mucosal immunity in crustaceans and provides insights for research on mucosal immunity in crustaceans.

1. Introduction

The immune system plays a crucial role in maintaining the physiological balance within an organism by recognizing and excluding foreign substances through immune responses. This system is broadly categorized into non-specific immunity and specific immunity [1]. Specific immunity, acquired through exposure to antigenic substances, primarily comprises cellular immunity mediated by T cells and humoral immunity mediated by B cells [2]. In contrast, innate immunity forms the foundation of specific immunity and includes immune barriers, intrinsic immune cells, and intrinsic immune molecules, which collectively provide rapid defense against most pathogens [3]. One key aspect is the immunological barrier inherent in the organism's physiological structure, such as respiratory, intestinal, and gastric mucosa, collectively referred to as the "first line of defense" against foreign substance entry [3–5]. Recent studies underscore the significance of mucosal immunity, a vital component of this immunological barrier, in resisting pathogen invasion [6]. For example, in mice, inflammatory bowel disease (IBD) is linked to dysfunctional mucosal immunity, with regulatory T cells, Th17 cells, and associated cytokines playing critical roles [7,8]. Mucosal immunity also defends against pathogenic *Candida albicans* through pathogen recognition and cytokine production [9]. Furthermore, mucosal immunity maintains intestinal homeostasis by defending against external pathogens and promoting tissue repair [10].

Mucosal immunity is present in various tissues with mucous membranes in vertebrates, including the intestines, stomach, genitourinary tract, respiratory tract, epithelium of exocrine glands, and conjunctiva of the eye [11]. This system, similar to cellular and humoral immunity, closely interacts with the lymphatic system to prevent potential pathogen invasion [12]. Acting as the primary interface with the external environment, the mucosal immune system regulates the organism's response to microorganisms on mucosal surfaces, preventing pathogens from entering the organism and causing infections [13]. Interestingly, comparative immunology reveals parallels between vertebrate mucosal immunity and invertebrate innate immune mechanisms [13]. Vertebrates, upon inflammation, experience the migration of leukocytes in the mucosa to the infection site, acting as a defense mechanism against microbial invasion [13]. In a similar vein, invertebrates like shrimp and crabs showcase a parallel defense mechanism, wherein hemocytes primarily migrate to the microbial infection area during inflammation, staunchly defending against pathogen invasion [14]. In vertebrates, mucosal tissues covering the digestive tract, such as the respiratory

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https://doi.org/10.1016/j.fsi.2024.109785
Received 16 April 2024; Received in revised form 10 July 2024; Accepted 20 July 2024
Available online 23 July 2024

mucosa, secrete mucus and interconnect with the lymphatic system to prevent pathogen invasion [15]. Similarly, in invertebrates like *Drosophila melanogaster*, an analogous defense mechanism exists in the form of the peritrophic matrix (PM), a layer of chitin and glycoproteins in the intestinal lumen acting as a physical and biochemical barrier [16,17]. Phylogenetically, mucous membranes appeared around 560 million years ago in the phylum cnidarians, like *Hydra*, exhibiting common functions with mucosal immunity in vertebrates [18]. Studies on the mammalian colon and *Hydra* suggest the presence of mucosal immunity in invertebrates [18–20]. Understanding mucosal immunity in economically important crustaceans, particularly shrimps and crabs, is crucial due to their role in marine biological cycles, economic significance, and potential impact on food safety and ecosystems [21–23]. As aquaculture develops, crustaceans such as crabs and lobsters have become vital and dominant components, driving major economic growth in many countries [24,25]. Therefore, crustaceans are often referred to as the "breadbasket of the sea." These organisms inhabit diverse environments globally and play central roles in ecosystem functions, controlling primary production and transferring energy and nutrients from primary producers to higher trophic levels [22]. In the context of global warming, enhancing oceanic carbon sinks is crucial for mitigating the greenhouse effect. Cultured organisms like crustaceans contribute to carbon sink fisheries by filtering large amounts of phytoplankton, facilitating carbon uptake from the waters [26,27].

Despite the economic and ecological importance of crustaceans, especially in the face of disease outbreaks, research on their immunity has predominantly focused on innate immunity, with limited attention to immune barriers [27]. This knowledge gap becomes particularly significant when considering the pivotal role crustaceans play in marine biological cycles, their economic importance in various industries, and the potential impact on food safety and ecosystems. Some immune defense mechanisms analogous to vertebrate mucosal immunity have been identified in crustaceans. For instance, a novel C-type lectin, MjGCTL, was discovered in the gill mucus of *Marsupenaeus japonicus*. In vivo, knockdown of MjGCTL impairs gill mucus agglutination and increases bacterial and *Vibrio* levels in the shrimp's gills and hemolymph [28]. However, the integrated mucosal immune defense mechanism of crustaceans remains unknown and deserves further exploration.

In this review, we delve into mucosal immunity in crustaceans, exploring mucosal tissues, immune cells, immune molecules, and host-microbial interactions. By addressing this research gap, we aim to enhance our understanding of how crustaceans protect themselves from pathogen infections. The insights gained not only contribute to the scientific understanding of emerging fields but also offer practical avenues for new species cultivation and strategies to bolster disease defense in aquaculture animals.

2. Mucosal tissues

Mucosal tissues, essential for orchestrating mucosal immunity, serve as the primary interface facilitating either direct or indirect contact with the external milieu of organisms [29]. This essential mucosal immune function is attributed to the mucous membrane covering tissue surfaces, which acts as a robust immune barrier, effectively preventing the entry of pathogens into the organism's internal environment [30]. Key mucosal immune tissues encompass the intestines, gills, stomach, and skin, each of which will be discussed below.

2.1. Intestines

The intestines, recognized as the largest immune organ in vertebrates, deploy lymphocytes and a mucous layer on the epithelial cells' surface to defend against bacterial invasion [31]. This robust defense mechanism is closely linked to the presence of gut-associated lymphoid tissue (GALT), a pivotal site for antigen sampling and the induction of adaptive immunity within the intestinal wall [32]. In murine models, GALT-produced IgA plays a crucial role in maintaining the symbiotic balance of gut flora [32]. In marine fish *Rachycentron canadum*, antigen-sensitized lymphocytes from GALT migrate via the bloodstream to mucosal tissues, forming mucosa-associated lymphoid tissue (MALT) [33]. Unlike mammalian GALT, teleost MALT primarily secretes IgT by B cells, which significantly contribute to intestinal mucosal immunity [34]. Consequently, mucosa, GALT/MALT, and their associated immunoglobulins constitute the fundamental elements of intestinal mucosal immunity in intestine. Further investigations in murine models have identified the forkhead box transcription factor (FOXO) protein as a regulator of B-cell development and immunoglobulin levels [35], potentially elucidating mechanisms regulating mucosal immunity. Intriguingly, in kuruma shrimp *Marsupenaeus japonicus* intestines, FOXO plays a pivotal role in maintaining the homeostasis of the intestinal flora [36]. Moreover, the intestinal lumen's surface, covered by a glycan structure and predominantly mucin (such as Muc2 in mice), forms a crucial protective layer [37]. Disruption in mucin synthesis or secretion alters the intestinal mucin layer's properties, increasing susceptibility to pathogens and disease [37,38].

Gel-forming mucins, produced by cuprocytes, represent essential macromolecules within the mucus layer of epithelial cells. Widespread and highly conserved in vertebrates and invertebrates [39], these mucins, exemplified by insect intestinal mucin (IIM) in *Trichoplusia ni*, confer protective functions against microbial infections [40]. Similar mucins are also present in arthropods, such as AgMuc1 in *Anopheles gambiae,* and LmIIM3 in *Locusta migratoria*, have been successively identified [41,42]. Crustaceans, such as shrimp *Penaeus vannamei*, hemocyanin acts as an important immune barrier in the hemolymph, which can be involved in the host's defense against the invasion of a variety of pathogenic bacteria. And the presence of hemocyanin in the intestinal mucus layer of *Penaeus vannamei* exhibit an intriguing association between hemocyanin and the intestinal mucus layer, hinting at a possible mucin-like role for shrimp hemocyanin [43,44]. Nonetheless, additional investigation is warranted to substantiate this hypothesis.

2.2. Gills

Gills are crucial for respiration, excretion, and other physiological processes, and they also act as entry points for opportunistic pathogens such as *Candidatus Piscichlamydia* and *Tenacibaculum maritimum* [45,46]. As a semi-permeable barrier between the organism and its external environment, gills represent the largest organ-specific surface interacting with external pathogens, including the filtration of pathogenic bacteria [45]. Evolutionarily, gills have developed mucosal immunity as the first immune barrier, particularly evident in fish [47]. In fish, such as amphioxus, gills have evolved into gas-exchange organs, forming a thin barrier between the fish's blood and the water environment to protect organisms from pathogen infection [45]. The mucus secreted by fish gills contains immunoglobulins, enzymes, antimicrobial peptides, and pathogen recognition receptors (PRRs) responding to pathogen-associated molecular patterns (PAMPs) [48,49]. Lectins, belonging to the humoral components of the innate immune system, play a crucial role in mucosal immunity. They can recognize PAMPs, influencing the agglutination of potentially pathogenic microorganisms or activating the complement components [50]. Therefore, the mucosal immunity of gills represents a comprehensive immune response process, encompassing recognition, complement system activation, and the release of antimicrobial peptides. In vertebrates like *Anguilla japonica*, C-type lectins isolated from gill mucus actively participate in host immune defense [51]. Moreover, in *Oreochromis niloticus*, rhamnose-binding lectins (RBLs) highly expressed in gills contribute to the innate immune response [52]. Despite the absence of adaptive immunity in invertebrates, lectins, and antimicrobial peptides in gills play crucial roles in innate immunity, recognizing and responding to pathogens [53,54]. The identification of a new antimicrobial peptide homolog, ApBD1, in the scallop *Argopecten purpuratus* further underscores the importance of gill epithelium in responding to pathogen stimulation [54]. In nematodes like *Caenorhabditis elegans*, over two hundred C-type lectins defend against pathogen invasion in innate immunity [55]. *Manila clam* exhibits six identified C-type lectins that may play roles in pathogen recognition and intercellular interactions [56]. Additionally, a new C-type lectin gene (MjGCTL) highly expressed in the gills of a crustacean, *Marsupenaeus japonicus*, suggests a gill defense mechanism akin to mucosal immunity [28]. This provides the first evidence of gill mucosal immunity in crustaceans, emphasizing the need for further exploration into the defense mechanisms in the gills of *Marsupenaeus japonicus*.

2.3. Other immune organs

In addition to the discussed immune tissues, mucosal functionality extends to organs like the stomach and skin [57,58]. The stomach, beyond digestion, actively participates in the organism's immune functions. In vertebrates, *Helicobacter pylori* infection leads to increased MALT, cell proliferation, and IgA production in the stomach of piglets, which in turn inhibits *Helicobacter pylori* value-added through mucosal immunity [59]. This suggests that the stomach, like other parts of the gastrointestinal tract, is capable of antigen processing and inducing mucosal and systemic immune responses [59]. In dromedary camels (*Camelus Dromedarius*), Peyer's patches (PPs) congregate within the gastrointestinal-associated lymphoid tissue (GALT) of the stomach, serving as pivotal sites for antigen presentation [60]. Despite the absence of dedicated mucosa-associated lymphoid tissue, the stomach harbors abundant antibody-producing plasma cells within the lamina propria (LP), notably generating immunoglobulin A to actively engage in mucosal immune responses [60]. Similarly, crustaceans demonstrate the multifunctional capacity of the stomach, fulfilling both digestive and immunological roles. In the crustacean *Marsupenaeus japonicus*, MjECSIT1, a homolog of ECSIT and a key component of the Toll signaling pathway, is identified in the stomach, where it performs a crucial antimicrobial function against gram-positive bacteria, similar to the bactericidal activity observed in mammalian macrophages [61].

Furthermore, the skin, acknowledged as a versatile tissue, not only maintains physiological equilibrium and guards against pathogenic bacteria but also serves as a prominent locus for mucosal immunity [62]. In vertebrates like rainbow trout, the skin's mucosa-associated lymphoid tissue (SALT) plays a crucial role in defending against invading pathogens [58]. Conversely, crustaceans have no skin tissue, but molting is a crucial biological process in crustaceans. Exposing vulnerabilities to infections, as observed in *Litopenaeus vannamei's* increased susceptibility to White Spot Syndrome Virus (WSSV) during this phase [63]. Additionally, the expression of pattern recognition proteins/receptors, such as C-type lectin receptor and Down's cell adhesion molecule homologs, significantly decreases during molting in *Portunus trituberculatu,* which further increases bacterial susceptibility [64]. Although skin associated with mucosal immunity in other animals, crustaceans lack skin tissue. These diverse tissues collectively provide immune cells, immune molecules, and nutrients to the mucosa/mucus layer, facilitating adaptive responses to fluctuating external environments.

3. Cells in the mucosal immune system

Cells serve as the fundamental building blocks for the constitution of tissues, playing a pivotal role in both structure and function. Within the realm of mucosal immunity, two primary cell types take center stage—epithelial and immune cells. Effective communication between these cells and their immediate surroundings is paramount for the maturation of the immune system.

3.1. Epithelial cells

Epithelial cells occupy the crucial interface between the tissue and the mucosal layer. Structurally, they exhibit a tightly arranged configuration, serving as a formidable barrier against the leakage of internal components into various body cavities. Functionally, these cells play a vital role in maintaining the homeostasis of mucosal immunity [65].

In vertebrates, particularly with in the intestinal context, epithelial cells orchestrate the symbiotic relationship between the host and the intestinal flora. This is accomplished through the construction of a robust mucosal barrier, secretion of diverse immune mediators, and delivery of bacterial antigens [66]. Notably, intestinal epithelial cells play a critical role under conditions like abdominal sepsis in mice by forming a chemical barrier that separates the intestinal lumen from the host's internal environment, actively participating in mucosal inflammatory and immune responses [67]. Interestingly, invertebrate epithelial cells show structural similarities, characterized by a tightly packed arrangement that aids in immune defense and nutrient absorption [68]. In invertebrate, these intestinal epithelial cells not only perform absorptive and secretory functions but also act as the first line of defense against pathogen infiltration [69,70]. The formation of cell sheets covering internal or external surfaces in invertebrates enables the regulation of substance passage between compartments. Investigating crustaceans unveils the versatile nature of epithelial cells, which play a role in decreasing metal contamination in edible seafood and mitigating foodborne pathogens like *Salmonella* and *Listeria monocytogenes*, thus lowering potential human health hazards [71]. In *Homarus americanus*, for example, epithelial cells facilitate metal uptake, whereas in *Litopenaeus vannamei*, they produce an anionic defense peptide called Stylicins, which defend against pathogen invasion [72,73]. This dual functionality encompasses not only the modulation of the host's antimicrobial immune response but also the facilitation of metallic element uptake through specific cation channels.

Crustacean epithelial cells share similarities with vertebrates in terms of nutrient absorption, water secretion, and pathogen defense. For example, in mammals such as humans, *Listeria monocytogenes* is a foodborne pathogen. In this context, intestinal epithelial cells serve as the first line of defense against *Listeria monocytogenes* by preventing bacterial adhesion to the epithelial lining [74]. Similarly, *Listeria monocytogenes* triggers an immune response through epithelial cells in snow crab, *Chionoecetes opilio*, to defend against bacterial invasion [74]. In summary, these observations not only emphasize the necessity for further exploration of their specific functions and mechanisms but also lay a potential theoretical groundwork for investigating how crustacean epithelial cells regulate and detoxify heavy metals in the environment, along with their potential role in drug resistance.

3.2. Immune cells

Epithelial cells serve as pivotal mediators in the communication between the microbiota and host immune cells, dynamically perceiving alterations in the luminal environment and initiating dialogues with immune cells in the underlying layers [75]. Subsequently, immune cells discern various signals within the microenvironment, thereby activating specific immune functions [76]. This diverse array encompasses lymphocytes, mast cells, and monocytes/macrophages, among others, with lymphocytes further classified into T cells and B cells. In vertebrates, mucosa-associated lymphoid tissues (MALT) house B cells, predominantly IgA-producing cells. Upon antigenic stimulation, these cells secrete IgA directly into nearby mucosa, thereby substantially contributing to mucosal immunity [15]. Mast cells (MCs), widely distributed around skin microvasculature and beneath mucous membranes [77], play a vital role in maintaining the intestinal barrier. They secrete various cytokines, modulating epithelial function and innate/adaptive defense responses [78]. Notably, certain granular blood cells in arthropods exhibit resemblances to mast cells, functioning as intermediaries between basophils and mast cells. They secrete histamine and heparin, which are recognized by pattern recognition receptors and binding to microbial cell walls [79].

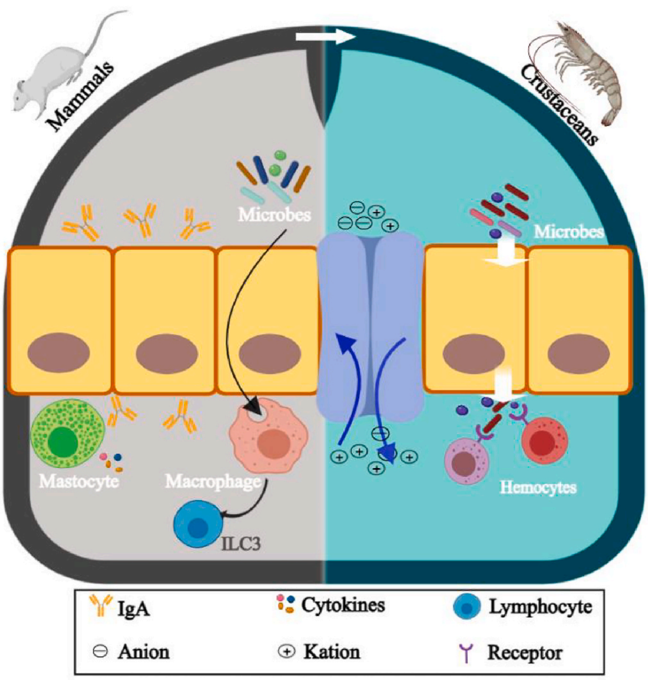

Fig. 1. Comparative insights into mucosal immune mechanisms in mammals and crustaceans. Intestinal epithelial cells in mammalian species secrete various immune mediators and deliver bacterial antigens by building mucosal barriers. Meanwhile, among immune cells, mast cells regulate innate and adaptive mucosal immunity by secreting a variety of cytokines. Macrophages can act as innate effector cells to phagocytose and clear bacteria, participate in the pathology of inflammatory bowel disease (IBD), and maintain host immunity and homeostasis in vivo. In crustaceans, intestinal epithelial cells apically possess specific transporter proteins that act on cations and anions. Also in immune cells, the ultrastructure of granulocytes is very close to that of mast cells, and blood cells exert immune action through receptor recognition and binding to microorganisms. In mammalian organisms, the intestinal epithelial cells play a pivotal role in the secretion of diverse immune mediators and the transportation of bacterial antigens through the construction of mucosal barriers. Concurrently, mast cells within the immune cell population modulate both innate and adaptive mucosal immunity via the secretion of a diverse array of cytokines. Macrophages serve as crucial innate effector cells, engaging in the phagocytosis and clearance of bacteria, contributing to the pathogenesis of inflammatory bowel disease (IBD), and preserving host immunity and homeostasis in vivo. Conversely, in crustaceans, the apical region of intestinal epithelial cells possesses specific transporter proteins that facilitate the transportation of cations and anions. Additionally, granulocytes in crustaceans exhibit an ultrastructure closely resembling that of mast cells, while hemocytes execute immune responses through the recognition and binding of microorganisms via receptor interactions.

Upon the breach of the intestinal barrier, monocytes initiate innate and adaptive immune responses, differentiating into macrophages [80]. Macrophages play a critical role in antimicrobial responses, serving as innate immune effectors that phagocytose and eliminate bacteria, secrete cytokines, and maintain the host's immune balance [80]. In vertebrates, macrophages regulate the mucosal barrier of intestinal epithelial cells, which is crucial for maintaining mucosal homeostasis and is involved in the pathogenesis of inflammatory bowel disease [65,80].

In crustaceans, hemocytes exhibit functional parallels with macrophages, depending on phagocytosis for the ingestion and eradication of bacteria, serving as a critical defense mechanism against various infectious agents such as bacteria and viruses [81,82]. For instance, the antimicrobial peptide Shcrustin in *Sinopotamon henanense* aids in clearing invading pathogens by binding to bacteria and promoting hemolymph phagocytosis [83]. The presence of mucus-associated hemocytes in the eastern oyster *Crassostrea virginica*, strategically located in pallial organs directly exposed to the marine environment, underscores their role as outposts. These hemocytes migrate bi-directionally between mucosal surfaces and the circulatory system, acting as immune cells defending against pathogen invasion through phagocytosis and influencing the immune system [84]. A comparative overview of mucosal immune mechanisms in mammals and crustaceans has been delineated in Fig. 1.

4. Mucosal Immune molecules

The regulation of mucosal immune responses depends on key immune molecules, which include immunoglobulins, antimicrobial peptides, and cell adhesion molecules. These molecules facilitate communication between immune cells, thereby regulating and controlling the mucosal immune response.

4.1. Immunoglobulins as Guardians of mucosal homeostasis

Immunoglobulins, as pivotal immune effectors, play a crucial role in local mucosal adaptive immune responses [85]. In mammals, IgA is the predominant isoform, mainly secreted at the intestinal mucosal surface. Its multifaceted functions include neutralizing toxins, combatting pathogenic microorganisms, and contributing to the maintenance of commensal flora homeostasis in the intestinal lumen [86,87]. In fish, however, only three immunoglobulins— IgM, IgD, and IgT—were found [88]. Among these, IgT plays a dominant role in fish mucosal immunity, protecting against invading microorganisms across mucous membranes of the gut, skin, and gills [88]. While IgM is the primary immunoglobulin isotype reacting against pathogens in systemic or mucosal compartments, its regulatory role in mucosal immunity is surpassed by IgT [89]. Additionally, IgD contributes to mucosal responses, primarily involving basophils in anti-microbial innate immunity [90].

In vertebrates, immunoglobulins are integral to mucosal immunity, but their absence in invertebrates is notable [91]. Interestingly, the presence of immunoglobulin superfamily (IgSF) molecules has been reported in invertebrates, suggesting a potential for mucosal immunity in this group [91]. For instance, *Biomphalaria glabrata* produces hemolymph polypeptides that belong to the IgSF and contain a fibronectin structural domain, enhancing defenses against pathogenic bacteria similarly to vertebrate immunoglobulin diversification [92]. In crustaceans infected with *Vibrio parahaemolyticus*, C-type lectin (*EsIg*Lectin), an Ig superfamily molecule, promotes bacterial phagocytosis within hemocytes through its Ig structural domain-dependent mechanism [93]. *EsIg*Lectin also enters also enters the intestinal mucosa to protect the gut from bacterial infection, thereby maintaining microbial homeostasis [93]. Similarly, in *Litopenaeus vannamei*, hemocyanin highly concentrated in the intestine exhibits an Ig-like protein with a C-terminal structural domain homologous to immunoglobulins [44], which recognizes bacteria such as *Vibrio parahaemolyticus* and triggers agglutination activity [44].

4.2. Immune signaling molecules

Both vertebrates and invertebrates exhibit a diverse array of immune signaling molecules aimed at eliminating pathogens or responding to local stimuli and stress. These molecules include antimicrobial peptides (AMPs) [94,95], and cell adhesion molecules like Dscam [96–98]. Immune signaling molecules are often closely associated with mucosal tissues and secretions. For instance, AMPs activate innate mucosal immune responses, playing a crucial role in eliminating pathogenic infections. Concurrently, cell adhesion molecules like Dscam, maintain endothelial cell integrity and are essential in the alternative adaptive immune system of invertebrate [99–101].

Throughout evolution, AMPs have emerged as ancient molecules crucial for bolstering the innate immune system's capacity to fend off invading pathogens. In pigs, these AMPs play a role in augmenting intestinal nutrient absorption, consequently enhancing mucosal immunity [102]. Furthermore, AMPs have been detected in the intestinal goblet cells of the grouper *Epinephelus coioides* and the mud crab *Scylla paramamosain*, indicating their potential secretion into the mucus layer [103,104].

Defensins, a significant family of AMPs, are widely found in both vertebrates and invertebrates, playing a crucial role in host defense against bacteria and fungi [105,106]. In triangle-shell pearl mussel *Hyriopsis cumingii*, six defensins contribute to the first line of immune defense against pathogen infection [107–109]. Similarly, the defensin-like peptide panusin in *Panulirus argus* exhibits anti-bactericidal activity akin to vertebrate β-defensins [110]. However, direct evidence of mucosal immunity involving AMPs in crustaceans remains elusive. In the absence of specific and adaptive immunity, invertebrates have evolved alternative strategies to combat various pathogens [97,111].

The Down syndrome cell adhesion molecule (Dscam) gene, with numerous spliced isoforms, plays a pivotal role in specific immune responses in invertebrates [96–98]. Dscam selectively recognizes bacteria and facilitates cellular phagocytosis [98,99]. In *Drosophila*, the utilization of Dscam transcripts contributes to bacterial phagocytosis [111]. Research on *Eriocheir sinensis* has demonstrated that Dscam generates a remarkable 30,600 isoforms, which have the ability to bind to a diverse array of bacteria [97]. The similarity in function between Dscam and antibodies, along with its process of selective splicing and the potential for immune memory observed in crayfish [112,113]. Consequently, these findings suggest the potential involvement of Dscam in mediating mucosal immunity in crustaceans, thereby providing a foundation for further investigations into its immunological functions.

5. Host-pathogen interactions at mucosal interfaces

The mucosal epithelium serves as a vital interface that mediates interactions between tissues and the external environment, playing a central role in host-pathogen interactions in both vertebrates and invertebrates. Upon encountering the microbiota, the mucosal epithelium orchestrates the formation of microvilli, mucus, and intercellular junctions. Infection by disease-causing microorganisms typically begins with their adherence to mucosal surfaces [20]. The abundant presence of carbohydrates and proteins in mucous secretions creates an optimal environment for microbial growth, thereby establishing a foundation for pathogenic invasion [114,115]. Noteworthy observations in *Acropora palmata* corals have revealed distinctions in bacterial communities between mucosal and aquatic environments, implying the mucus' role in recruiting and maintaining specific microorganisms [116].

Examination of vertebrate mucus, particularly in fish, reveals a complex composition featuring hydrophobic and antimicrobial proteins that regulate the growth of various bacterial species [117,118]. Glycoproteins and peptidoglycans within mucus facilitate interactions between sugars and proteins, leading to the binding of specific mucosal ligands via adhesins. This interaction fanumerous advantages [119]. Remarkably, microorganisms possess the ability to influence mucosal dynamics. Both commensal and pathogenic microorganisms have been identified as capable of attaching to and degrading mucous layers [31]. For instance, cholera toxin from *Vibrio cholerae* triggers significant mucin secretion and release in mammals like mice through a cAMP-dependent mechanism [120,121]. Conversely, certain infectious agents, such as *C. difficile* in humans, have been shown to inhibit mucus secretion [122]. These findings underscore the intricate mechanisms through which microorganisms modulate host responses by mediating mucous secretion, potentially facilitating pathogen invasion. In summary, the production and composition of mucus directly impact microbial colonization, while microorganisms, in turn, can alter and regulate mucous secretion. These reciprocal interactions between microbes and mucus-producing cells highlight the nuanced interplay at the host-pathogen interface [31].

6. Factors affecting immune response of crustaceans

Crustaceans live in diverse aquatic environments and are influenced by a variety of abiotic and biotic factors that significantly impact their immune responses. Abiotic factors such as temperature, pH, salinity, dissolved oxygen, and ammonia are particularly important in modulating crustacean immunity.

6.1. Abiotic factors

Temperature fluctuations profoundly affect crustacean immune parameters. High temperatures can induce apoptosis of hemocytes and reduce phenoloxidase activity, thereby increasing susceptibility to pathogens [123]. Conversely, low temperatures decrease mitochondrial aerobic capacity and alter immune responses such as total hemocyte count and phagocytosis [124,125]. For instance, temperature variation in signal crayfish showed differential immune responses and disease susceptibility [126]. pH variations in crustacean habitats impact their immune competence. Both high and low pH levels can suppress immune functions, including phenoloxidase activity and phagocytosis, affecting resistance to pathogens like *Vibrio alginolyticus* [127,128].

pH fluctuations also influence bacterial virulence, altering disease dynamics in crustacean hosts [129,130]. Drastic changes in salinity disrupt osmoregulation and compromise crustacean immune systems, exacerbating pathogen virulence and mortality rates [131,132]. Dissolved oxygen fluctuations, often due to environmental stressors, trigger hypoxia-inducible factors that modulate immune responses in species such as *Litopenaeus vannamei* [133,134]. High ammonia levels impair ion regulation and immune functions in crustaceans [135–137]. In vertebrates like fish, changes in temperature, pH, and dissolved oxygen have been shown to affect the microbiota of mucosal surfaces, thereby altering mucosal immunity [138]. Therefore, we hypothesized that environmental factors such as temperature and pH may also affect mucosal immunity in crustaceans.

6.2. Biotic factors

Pathogenic challenges significantly alter crustacean immune indices. For instance, infections like acute hepatopancreatic necrosis disease (AHPND) activate pattern recognition receptors, enhancing antimicrobial activities in shrimp [139]. Infections with various pathogens induce differential responses in immune enzymes and proteins such as pro-phenoloxidase and lysozyme, indicating pathogen-specific immune interactions [140–142].

7. Conclusion

Our review asserts that invertebrates possess not only "immune memory" but also a nuanced "mucosal immunity." This revelation, exemplified in crustaceans like *Marsupenaeus japonicus*, expands our comprehension of the immune landscape in invertebrates. Furthermore, insights from comparative immunology unveil a remarkable convergence between vertebrate mucosal immunity and the innate immunity mechanisms observed in invertebrates [13]. For instance, the peritrophic matrix (PM) in *Drosophila melanogaster* exhibits striking similarities to vertebrate mucosal tissue, housing mucins, and chitin. This structural parallel serves as an immune barrier, shielding host cells from microbial invasion [143].

These compelling findings not only underscore the existence of mucosal immunity in crustaceans but also highlight the evolutionary conservatism shared between vertebrates and invertebrates. Noteworthy is the resemblance of the peritrophic matrix in *Drosophila melanogaster* to mucosal tissue in vertebrates, emphasizing a shared protective function against microbial intrusion. Such revelations pave

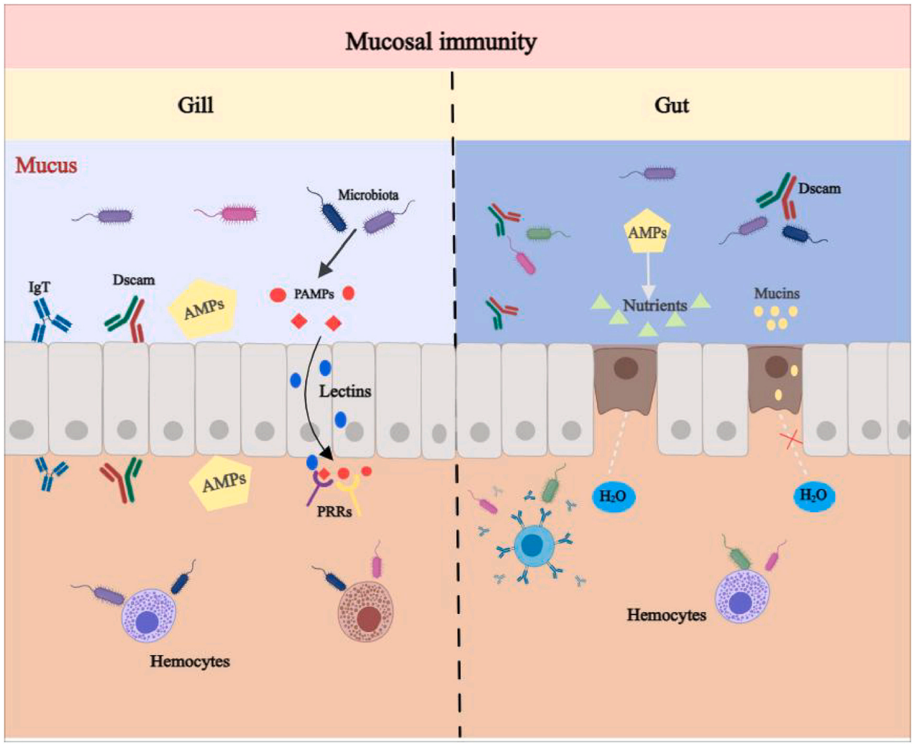

Fig. 2. A schematic diagram of mucosal immune mechanisms in crustaceans. In crustaceans, the gill serves as the primary immune organ protecting the organism against pathogen invasion. Within gill mucus, there are immunoglobulins (Igs), antimicrobial peptides, and pathogen recognition receptors (PRRs). Among these components, PRRs play a pivotal role in monitoring the mucosal surface response to pathogen-associated molecular patterns (PAMPs). Lectins are implicated in PAMP recognition, influencing the agglutination of potentially pathogenic microorganisms and the activation of complement components. The intestine, recognized as the largest immune organ in the body, is surrounded by microbiota. Mucins within the mucus layer of the intestinal lumen contribute to maintaining the mucosal barrier due to the water retention capacity conferred by the glycan structure of their surface. Antimicrobial peptides in the gut facilitate nutrient absorption and water secretion by intestinal epithelial cells, thus bolstering the immunity of the intestinal mucosa. Down syndrome cell adhesion molecules (Dscam) in the intestine share structural and functional similarities with antibodies and play a role in recognizing various types of bacteria, promoting cell phagocytosis.

the way for extrapolating insights into crustacean mucosal immunity from other species, providing a basis for hypothesized efficacy in crustacean aquaculture.

In conclusion, this comprehensive review delves into the intricate landscape of crustacean mucosal immunity, shedding light on the crucial contributions of various tissues, cells, and molecules in the defense against pathogens [144]. Mucosal tissues, acting as primary barriers to external threats, play a pivotal role in thwarting pathogen invasion, with a specific focus on the indispensable role of gel-forming mucins in gut mucosal immunity. The examination of gill mucosal immunity, reminiscent of fish mechanisms, highlights the involvement of lectins and antimicrobial peptides, showcasing its significance in respiratory and excretory processes. The exploration of immune cells, particularly macrophages, and hemocytes, underscores their pivotal role in bacteria clearance, paralleling antimicrobial responses observed in both crustaceans and vertebrates. The proposed mechanistic map emphasizes the interconnectedness of mucosal immune cells in maintaining internal homeostasis (Fig. 2). Analysis of mucosal immune molecules, such as immunoglobulins, antimicrobial peptides, and cell adhesion molecules, unveils alternative strategies in crustaceans, including the presence of Ig superfamily molecules and Ig-like domains in hemocyanin. Additionally, this review underscores the importance of immune signaling molecules, such as antimicrobial peptides and Dscam, showcasing the diverse strategies used in crustacean mucosal immunity. By emphasizing the pivotal role of the mucosal epithelium in host-pathogen interactions, it advances our understanding of crustacean mucosal immunity. These insights not only enhance our knowledge of emerging fields but also provide practical approaches for species cultivation and disease management in aquaculture. Ultimately, this foundational platform deepens comprehension and encourages further research into crustacean mucosal immunity, which is essential for maintaining health and productivity in aquaculture under environmental pressures.

CRediT authorship contribution statement

Ruixue Hao: Conceptualization, Data curation, Formal analysis, Software, Visualization, Roles/Writing-original draft. Mingming Zhao: Conceptualization, Data curation, Formal analysis. Muhammad Tayyab: Data curation, Formal analysis, Supervision, Writing-review & editing. Zhongyang Lin: Conceptualization, Supervision, Funding acquisition, Writing-review & editing. Yueling Zhang: Conceptualization, Funding acquisition, Project administration, Supervision, Writing-review & editing.

Declaration of Competing Interest

The authors declare no conflict of interests.

Data availability

No data was used for the research described in the article.

Acknowledgments

This work was supported by grants from National Natural Science Foundation of China [Grant number U22A20536] and STU Scientific Research Initiation Grant [Grant number NTF23001].

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